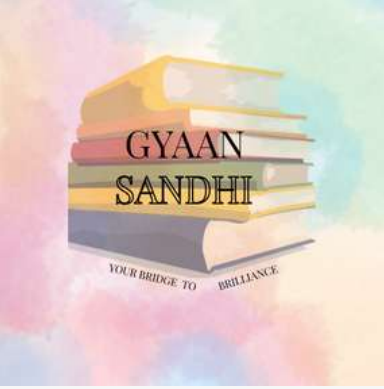




Information Security

Unit 6

Cyber Security and Tools

Cyber Crime

Cyber crime refers to illegal activities performed using computers, networks, or the internet. It includes crimes such as hacking, phishing, identity theft, online fraud, cyber stalking, malware attacks, and data theft. Cyber criminals target individuals, organizations, and governments to steal sensitive information, damage systems, or gain financial benefits. With the rapid growth of digital technology and online services, cyber crimes have increased significantly across the world. These attacks can lead to financial losses, privacy violations, and disruption of important services. Preventive measures such as strong passwords, antivirus software, firewalls, encryption, and user awareness help reduce cyber crime risks.

Information Security

Information security is the practice of protecting information and computer systems from unauthorized access, misuse, disclosure, disruption, modification, or destruction. The main objective of information security is to ensure confidentiality, integrity, and availability of data, commonly known as the CIA triad. It involves the use of security policies, encryption techniques, authentication methods, firewalls, and access controls to secure information. Information security is important for protecting personal data, business records, financial transactions, and government information from cyber threats. Effective information security helps maintain trust, privacy, and smooth functioning of digital systems and communication networks.

Cyber Terrorism

- Cyber terrorism refers to the use of computers, networks, and the internet to carry out terrorist activities that create fear, panic, or disruption in society.
- It involves attacks on critical systems such as government networks, banking systems, power grids, transportation systems, and communication networks. Cyber terrorists use techniques like hacking, spreading malware, launching denial-of-service attacks, and stealing sensitive information to damage national security and public services.
- The main objective of cyber terrorism is to threaten governments, spread fear among people, and cause economic or social instability. Due to increasing dependence on digital technology, cyber terrorism has become a major global security concern.

Cyber Terrorism

- Due to increasing dependence on digital technology, cyber terrorism has become a major global security concern. Strong cybersecurity measures, continuous monitoring, awareness programs, and international cooperation are important to prevent cyber terrorism attacks.

Cyber Terrorism in India

- Cyber terrorism in India involves the use of computer networks and the internet to threaten national security, public safety, and critical infrastructure.
- India has faced cyber attacks targeting government websites, banking systems, defense networks, railway services, and communication systems.
- Terrorist groups and cyber criminals use hacking, malware, phishing, and denial-of-service attacks to disrupt online services and steal sensitive information.
- Critical sectors such as power grids, healthcare, transportation, and financial institutions are major targets of cyber terrorism.
- Social media and online platforms are sometimes misused to spread fake news, propaganda, fear, and extremist ideologies.

Cyber Terrorism in India

- The increasing use of digital services and online transactions has raised cybersecurity challenges in India.
- The Indian government has established organizations like CERT-In (Computer Emergency Response Team – India) to monitor and respond to cyber threats.
- Laws such as the Information Technology Act 2000 provide legal measures to control cyber crimes and cyber terrorism.
- India promotes cybersecurity awareness, digital security training, and strong data protection measures to reduce cyber terrorism risks.
- International cooperation and advanced cybersecurity technologies are important for protecting India from cyber terrorist attacks.

Cyber Crime Against Property

- Cyber crime against property refers to illegal activities in which computers, digital devices, networks, or online data are damaged, stolen, misused, or destroyed. In these crimes, the target is the property or data of an individual, organization, or government. Cyber criminals use unauthorized access, malware, hacking, and other techniques to harm digital assets and intellectual property. These crimes can result in financial losses, data corruption, and disruption of business operations.

Examples of Cyber Crime Against Property

- Hacking into computer systems and stealing confidential data.
- Spreading viruses, worms, and malware to damage files and software.
- Data theft and unauthorized copying of important information.
- Intellectual property theft such as software piracy and copyright violations.
- Cyber vandalism that damages websites or online databases.

Cyber Crime Against Property

- Online financial fraud and unauthorized fund transfers.
- Destroying or modifying digital records and databases.
- Stealing passwords, digital documents, or business information.

Prevention Measures

- Use strong passwords and authentication methods.
- Install antivirus software and firewalls.
- Regularly update software and security systems.
- Encrypt sensitive data and maintain backups.
- Avoid downloading files from untrusted sources.
- Conduct cybersecurity awareness and training programs.

Types of Cyber Crimes

1

HACKING



Unauthorized access to computer systems or networks to steal, alter or destroy data.

2

THEFT



Stealing money, data, or digital resources through computers and the internet.

3

CYBER STALKING



Using the internet or digital platforms to harass, threaten or repeatedly monitor a person.

4

IDENTITY THEFT



Stealing personal information like passwords, ID details or bank information to commit fraud.

5

MALICIOUS SOFTWARE



Creating and spreading harmful software like viruses, worms, Trojans to damage systems or steal data.

6

CHILD SOLICITING



Using the internet to contact, communicate or entice children for illegal or inappropriate purposes.

7

CHILD ABUSE



Exploiting and abusing children through the internet, including sharing or distributing abusive content.

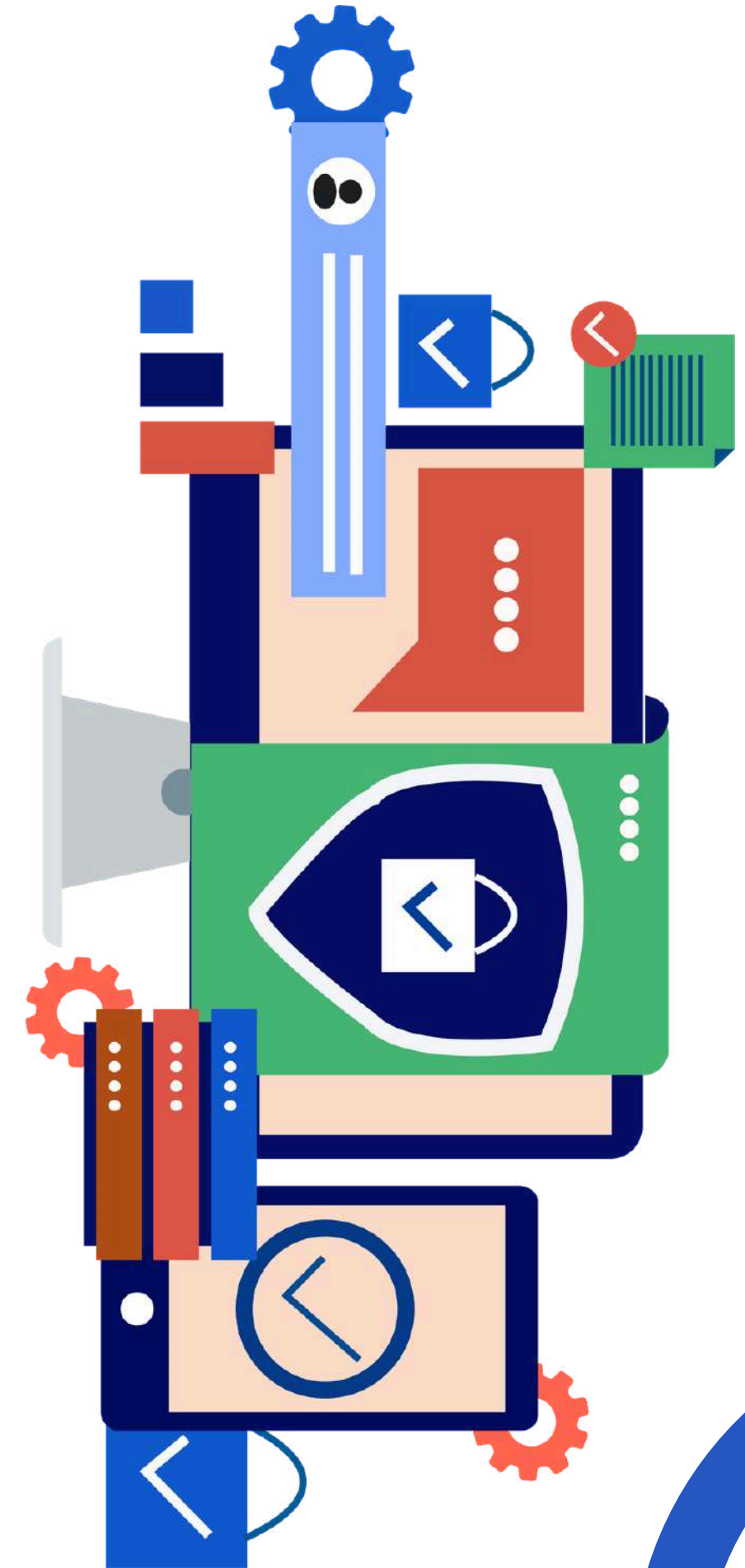


Stay Safe Online – Be Aware, Be Smart, Be Secure.



Botnets

- A botnet is a group of computers, mobile devices, or internet-connected systems that are infected with malicious software and controlled remotely by a cyber criminal, known as a botmaster. The infected devices work together as a network to perform illegal or harmful activities without the knowledge of the device owners. Each infected device in the network is called a bot or zombie.
- Cyber criminals create botnets by spreading malware through phishing emails, unsafe downloads, infected websites, or software vulnerabilities. Once the malware enters the device, the attacker gains control over it through a command-and-control (C&C) server. The attacker can then send instructions to thousands of infected devices simultaneously.

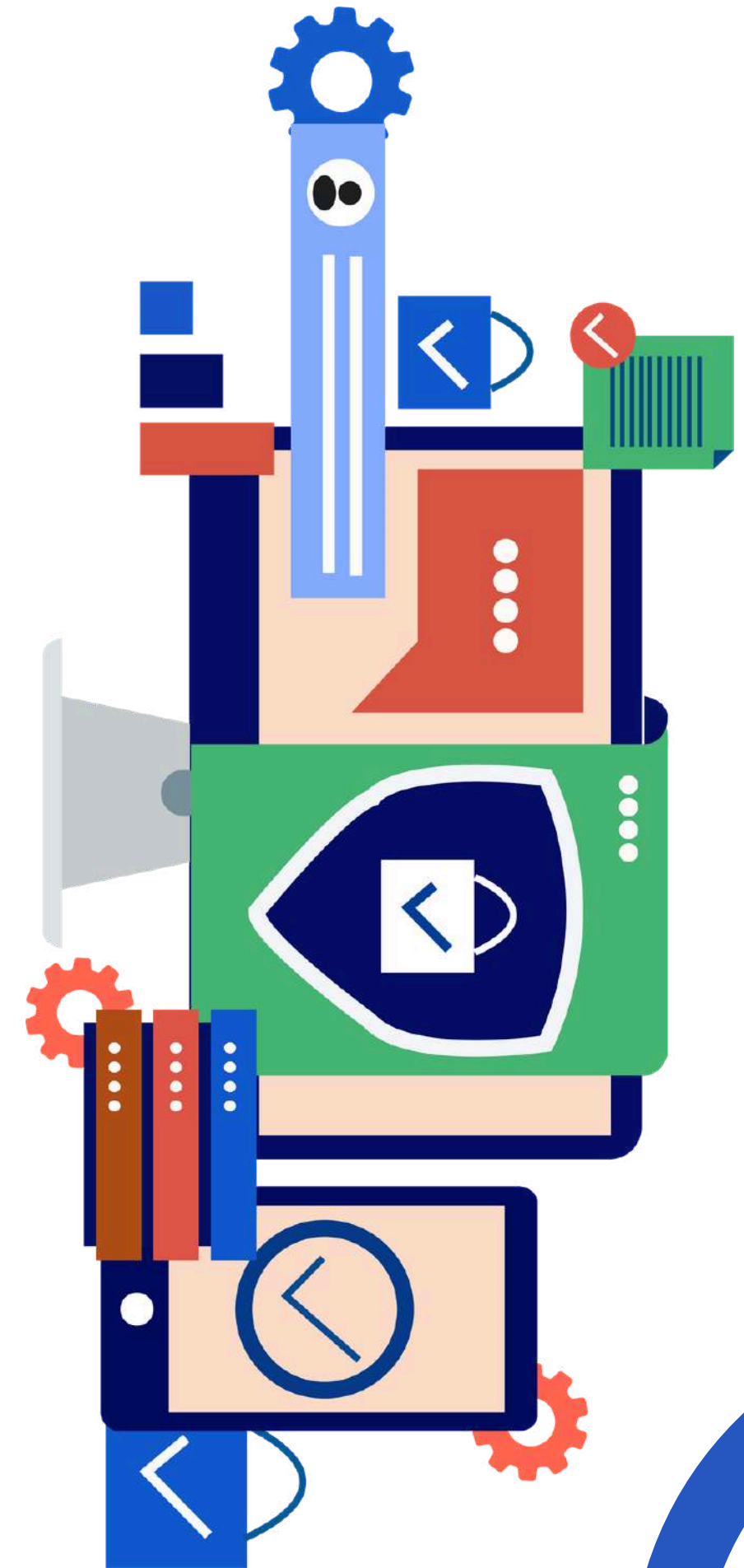


Botnets

- Botnets are mainly used for Distributed Denial of Service (DDoS) attacks, where a large number of devices flood a server or website with traffic, causing it to crash or become unavailable. They are also used for sending spam emails, stealing confidential information, cryptocurrency mining, spreading malware, and online fraud.

Features of Botnets

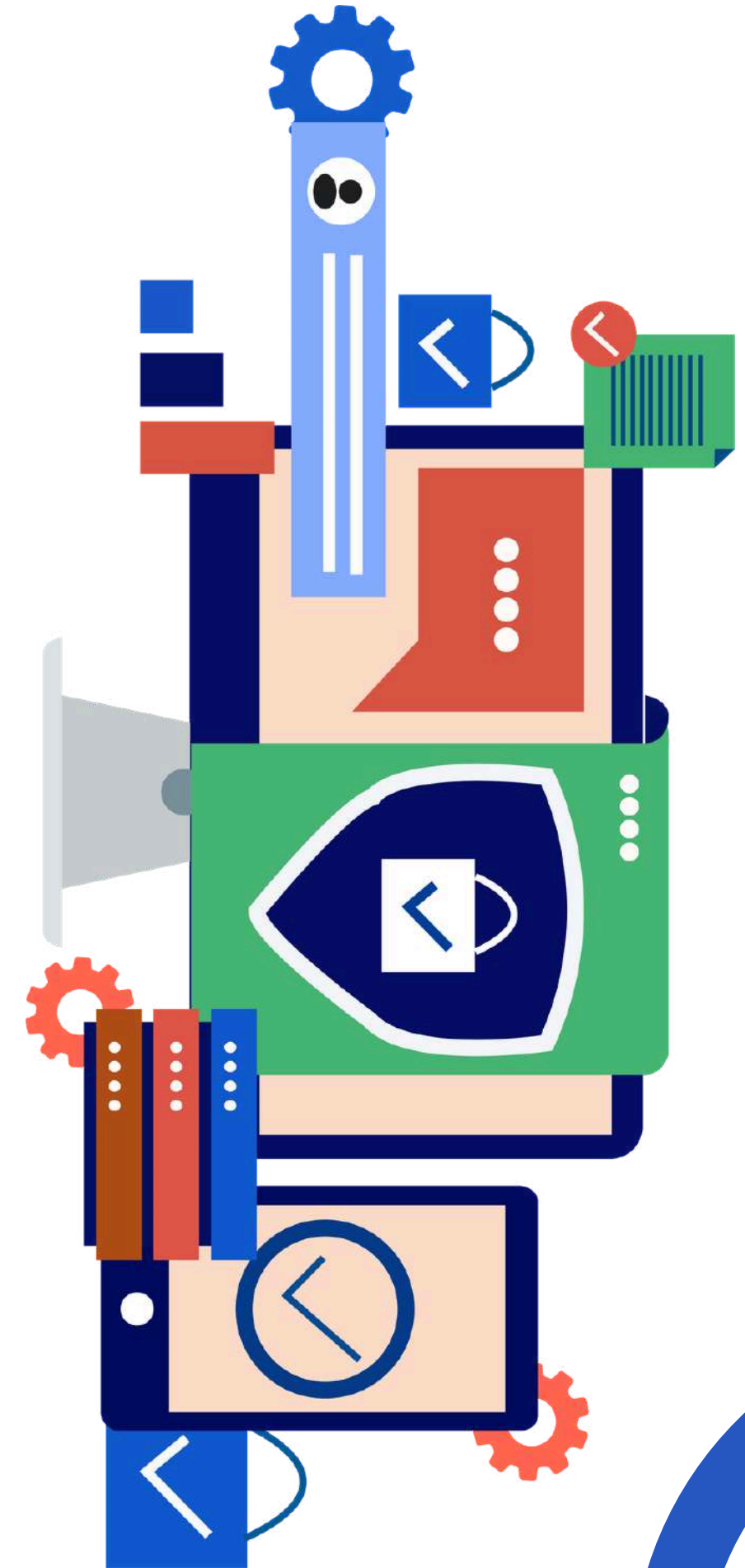
- Large network of infected devices
- Controlled remotely by hackers
- Operate without the user's knowledge
- Used for large-scale cyber attacks
- Difficult to detect and stop



Botnets

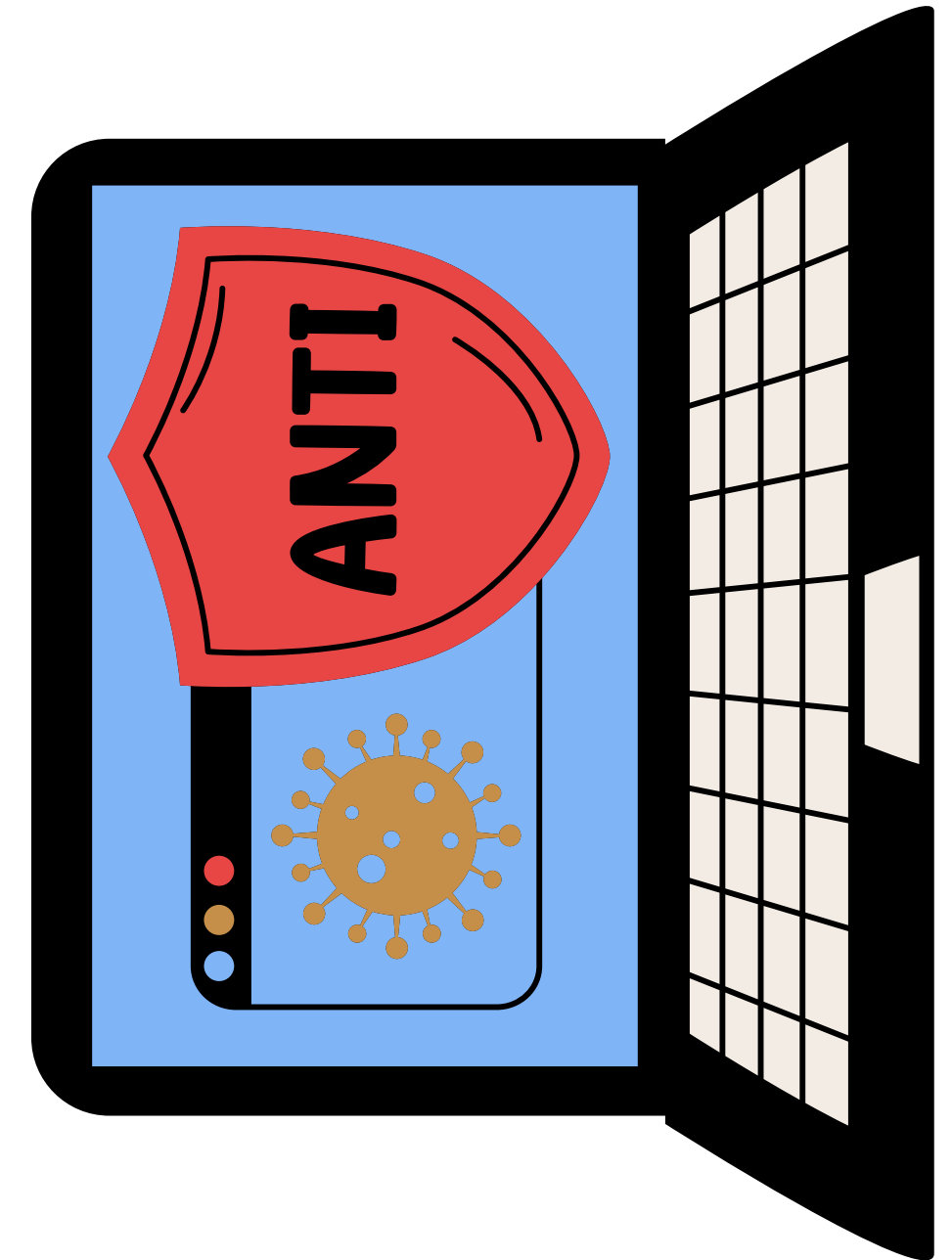
Effects of Botnets

- Slow internet and system performance
- Data theft and privacy loss
- Financial damage to organizations
- Website and server crashes
- Spread of malware across networks



Zombies

- A zombie is a computer or device that has been infected with malware and secretly controlled by a hacker. The owner of the device usually does not know that their system is compromised. Zombies become part of a botnet and carry out malicious activities based on commands received from the botmaster.
- Hackers use zombies to increase the power of cyber attacks. A single zombie may not cause much damage, but thousands of zombies working together can create powerful attacks. Zombies are often used to send spam messages, spread viruses, attack websites, or steal sensitive data.

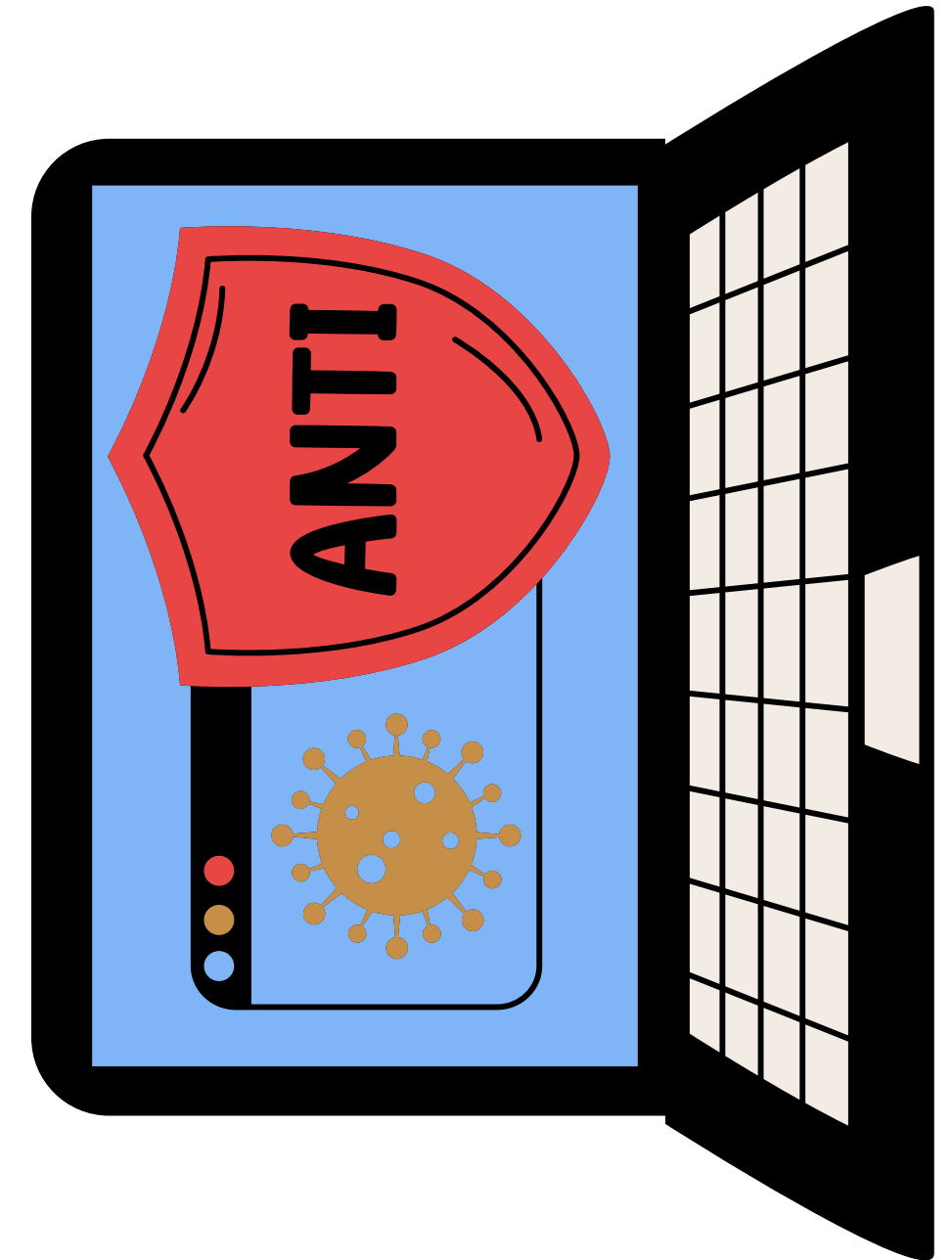


Zombies

- A zombie device may show signs such as slow performance, unusual internet activity, unexpected pop-ups, overheating, or unknown programs running in the background. Proper cybersecurity measures help prevent systems from becoming zombies.

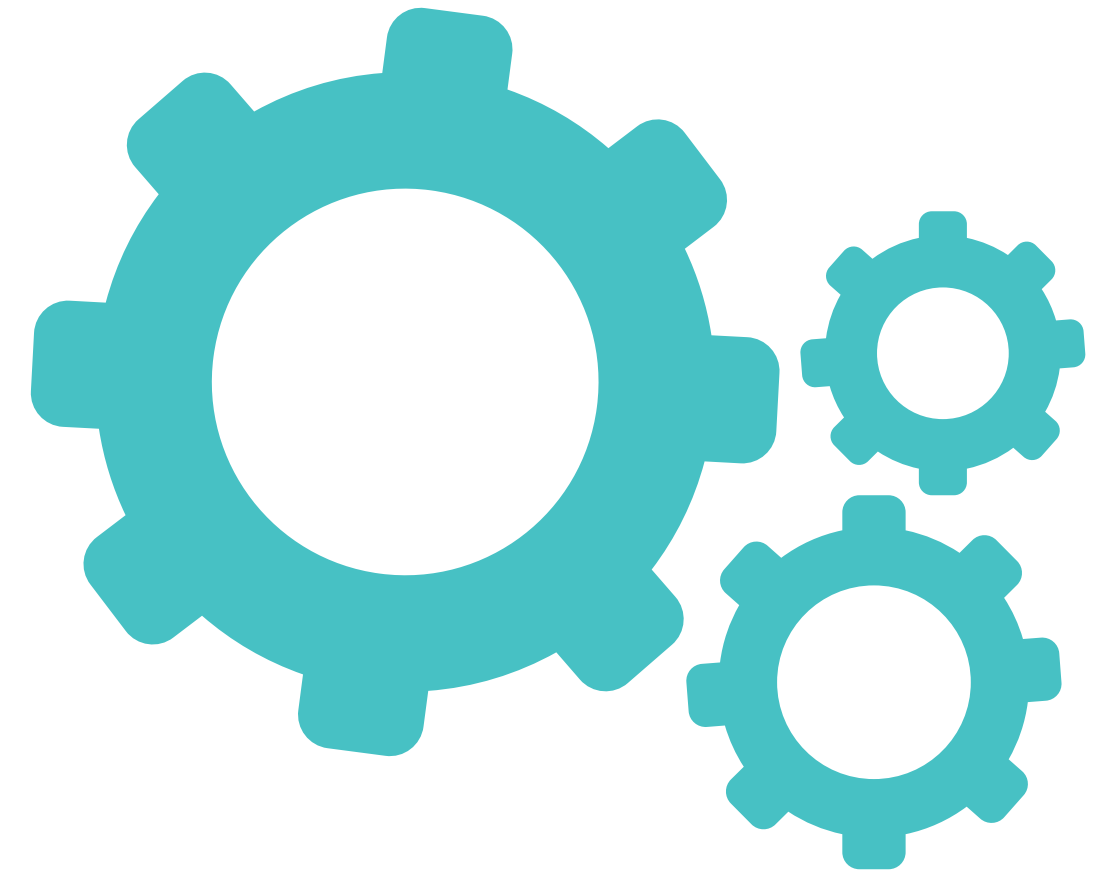
Prevention Measures

- Install updated antivirus and anti-malware software
- Use firewalls and secure networks
- Avoid suspicious emails and unknown downloads
- Regularly update operating systems and applications
- Use strong passwords and multi-factor authentication
- Scan systems regularly for malware infections



Social Engineering

- Social engineering is a cyber attack technique in which attackers manipulate, deceive, or trick people into revealing confidential information or performing actions that compromise security. Instead of attacking computer systems directly, cyber criminals target human behavior and emotions such as fear, trust, curiosity, or urgency to gain access to sensitive data like passwords, bank details, or personal information.
- Social engineering attacks are commonly carried out through emails, phone calls, social media, or fake websites. Attackers pretend to be trusted persons or organizations to fool victims. These attacks are dangerous because they exploit human mistakes rather than technical weaknesses.



Social Engineering

Common Types of Social Engineering

1. Phishing

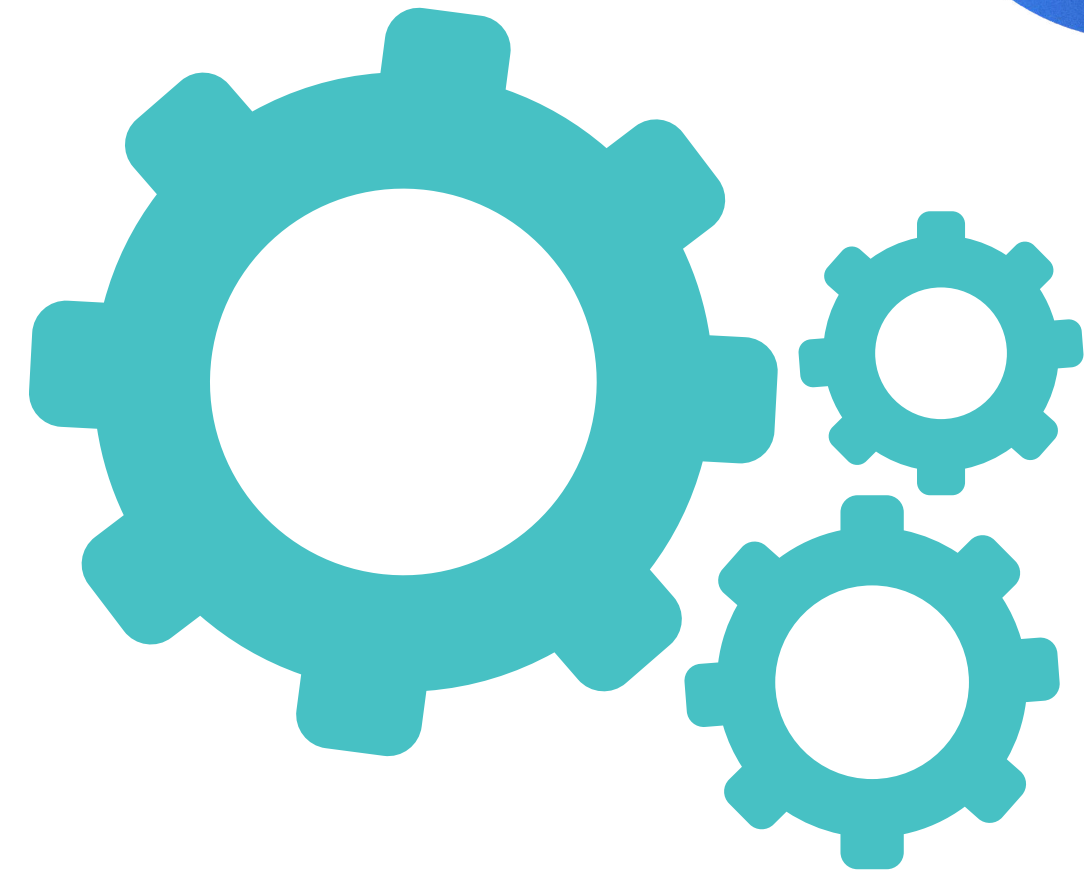
- Phishing involves sending fake emails or messages that appear to come from trusted organizations such as banks or companies. The goal is to steal usernames, passwords, or financial details.

2. Pretexting

- In pretexting, attackers create a fake identity or story to gain the victim's trust and collect confidential information.

3. Baiting

- Baiting tricks users by offering something attractive such as free software, movies, or USB drives that contain malware.



Social Engineering

4. Tailgating

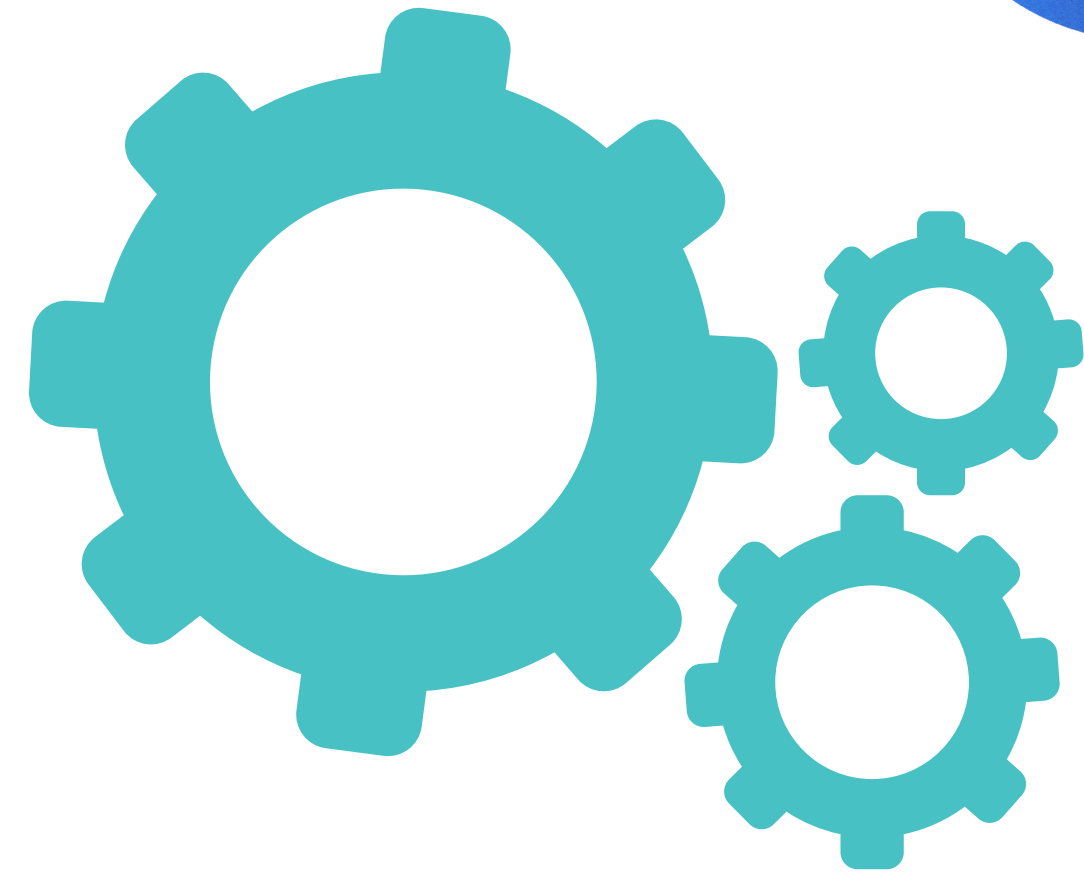
Tailgating occurs when an unauthorized person gains physical access to a secure area by following an authorized person.

5. Scareware

Scareware uses fake warnings or security alerts to frighten users into installing malicious software or paying money.

Effects of Social Engineering

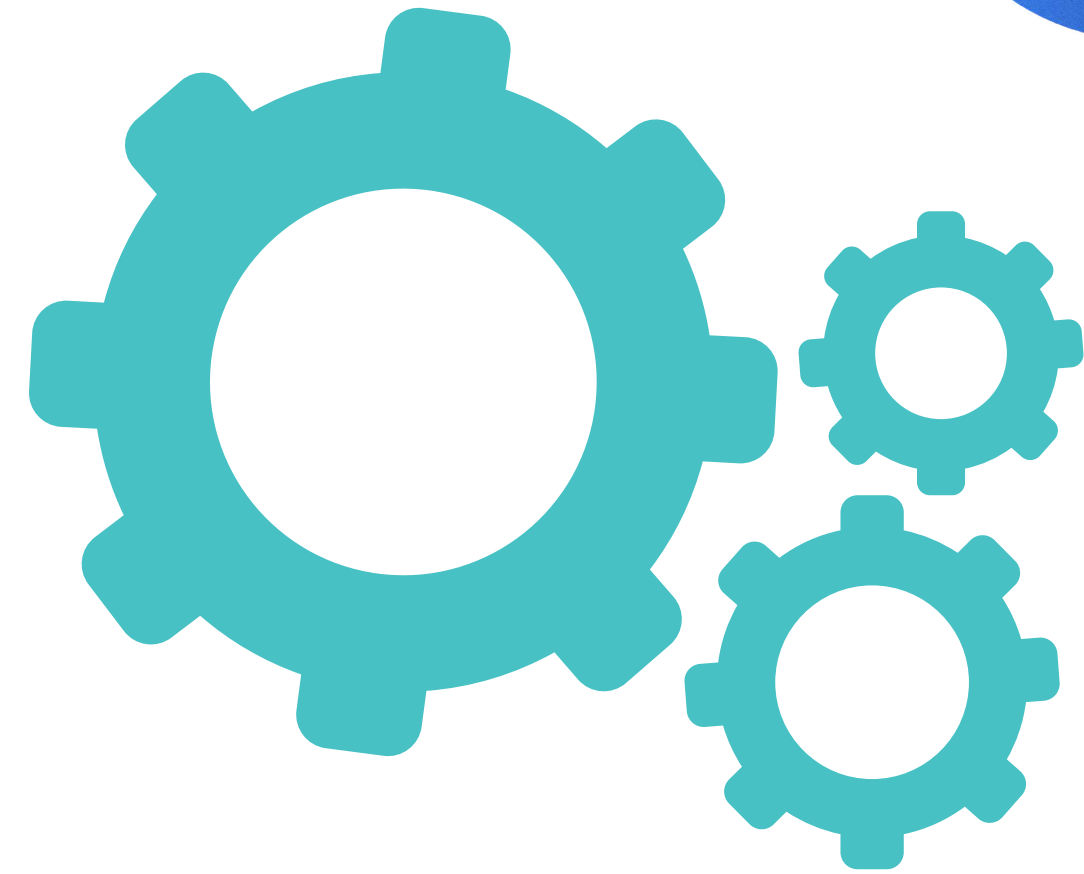
- Theft of personal and financial information
- Unauthorized access to systems
- Financial loss
- Data breaches
- Identity theft



Social Engineering

Prevention Measures

- Avoid clicking suspicious links or attachments
- Verify the identity of unknown callers or emails
- Use strong passwords and multi-factor authentication
- Keep software and security systems updated
- Provide cybersecurity awareness training
- Never share confidential information with unknown sources



Cyber Stalking

CYBER STALKING

When Online Harassment Follows You Everywhere

Cyber stalking is the repeated use of the internet, social media, emails or other digital platforms to **harass, threaten, monitor** or **frighten** a person. It is a serious cyber crime that affects the victim mentally, emotionally and sometimes physically.

WHAT IS CYBER STALKING?

It is when someone uses digital means to repeatedly threaten, harass, track, or contact you without your consent, causing fear or distress.



- Threatening Emails
- Abusive Messages
- Fake Profiles
- Tracking Location



EFFECTS OF CYBER STALKING

- Mental stress, anxiety and fear
- Loss of privacy and confidence
- Emotional harassment
- Damage to reputation
- Financial and personal problems

COMMON METHODS OF CYBER STALKING

-  Sending threatening or abusive emails and messages
-  Monitoring social media activities
-  Creating fake profiles to harass victims
-  Spreading false rumors or personal information online
-  Tracking a person's online behavior and location
-  Hacking accounts to collect private information

PREVENTION MEASURES

- Keep your social media accounts private.
- Avoid sharing personal information online.
- Use strong passwords and two-factor authentication.
- Block and report suspicious users.
- Save evidence of threatening messages.
- Use updated antivirus and security software.





LEGAL ACTION

In India, cyber stalking is punishable under the Information Technology Act, 2000 (Section 66E) and other cyber crime laws. Victims can report such incidents to:



Cyber Crime Helpline Number 1930



Online Cyber Crime Reporting Portal www.cybercrime.gov.in



Nearest Cyber Crime Police Station

Type of Stalkers

1. Simple Obsessional Stalker

A simple obsessional stalker is a person who already knows the victim, such as a former friend, partner, colleague, or acquaintance. This type of stalker becomes obsessed after a conflict, breakup, or rejection and repeatedly tries to contact, threaten, or harass the victim. They may send continuous messages, monitor activities, or try to control the victim's life.

2. Delusional Stalker

A delusional stalker believes that the victim loves or admires them even when it is not true. This stalker often creates imaginary relationships and may think they have a special connection with the victim. Delusional stalkers usually target celebrities, public figures, or strangers and continue stalking despite rejection.

Type of Stalkers

3. Vengeful Stalker

A vengeful stalker targets a person to take revenge or create fear and emotional pain. Their main objective is not love or attention but to punish or intimidate the victim. They may spread false rumors, send threats, damage reputation, or perform online harassment to disturb the victim mentally and emotionally.



How to Identify and Detect Cyber Stalking

- Receiving repeated unwanted emails, messages, or phone calls from the same person.
- Getting threatening, abusive, or disturbing online messages.
- Someone constantly monitoring your social media activities and posts.
- Fake profiles being created in your name or used to contact you.
- Personal information or photos being shared online without permission.
- Unknown persons trying to track your location or online activity.
- Frequent hacking attempts on your accounts or password reset notifications.
- Receiving messages from multiple fake or anonymous accounts.
- Sudden appearance of spyware or suspicious applications on devices.
- Feeling continuously watched, harassed, or followed online.

How to Identify and Detect Cyber Stalking

Methods to Detect Cyber Stalking

- Check account login history and suspicious device activity.
- Monitor unusual emails, links, or attachments.
- Use antivirus and anti-spyware software to scan devices.
- Review privacy settings on social media accounts.
- Keep records and screenshots of threatening messages.
- Observe unusual internet or system behavior such as slow performance or unknown apps.
- Enable two-factor authentication for online accounts.
- Report suspicious activities to cyber crime authorities or online platforms.



Proxy Servers

A proxy server is an intermediary server that acts as a bridge between a user's computer and the internet. When a user requests a webpage or online service, the request first goes to the proxy server, which then forwards it to the destination server. The response from the website is sent back through the proxy server to the user. Proxy servers help improve security, privacy, and network performance.

Proxy servers are commonly used in organizations, schools, and businesses to control internet usage, filter content, and protect internal networks from cyber threats. They can also hide the user's IP address, making online activities more private and secure.

Functions of Proxy Servers

- Hides the user's IP address
- Improves network security
- Filters unwanted or harmful websites

Proxy Servers

- Controls internet access
- Stores cached webpages for faster access
- Monitors and logs internet activity

Types of Proxy Servers

1. Forward Proxy

A forward proxy works on behalf of the client and forwards user requests to the internet. It is commonly used for privacy, filtering, and access control.

2. Reverse Proxy

A reverse proxy works on behalf of a web server. It receives requests from users and forwards them to the appropriate server. It improves security, load balancing, and performance.

Proxy Servers

3. Anonymous Proxy

- An anonymous proxy hides the user's identity and IP address while accessing websites.

4. Transparent Proxy

- A transparent proxy does not hide the user's identity and is mainly used for monitoring and filtering internet traffic.

Advantages of Proxy Servers

- Enhances privacy and security
- Reduces bandwidth usage
- Speeds up web access through caching
- Blocks malicious websites
- Helps in content filtering

Anonymizers

An anonymizer is a tool or service that helps users hide their identity and online activities while browsing the internet. It acts as an intermediary between the user and the website, making it difficult to trace the user's real IP address and location. Anonymizers are mainly used to maintain privacy, protect personal information, and access blocked or restricted websites.

Anonymizers work by routing internet traffic through proxy servers or encrypted networks. They help users browse anonymously by masking identifying details from websites, advertisers, or cyber attackers. While anonymizers provide privacy and security benefits, they can also be misused for illegal activities.

Features of Anonymizers

- Hide the user's IP address
- Protect online privacy

Anonymizers

- Encrypt internet traffic
- Prevent tracking by websites
- Allow anonymous browsing
- Help access restricted content

Types of Anonymizers

1. Proxy-Based Anonymizers

These anonymizers use proxy servers to hide the user's identity and forward requests to websites.

2. VPN (Virtual Private Network)

A VPN encrypts internet traffic and hides the user's IP address by routing data through secure servers.

Anonymizers

3. Onion Routing

- This method uses multiple layers of encryption and routes traffic through several servers to provide high anonymity.

Advantages of Anonymizers

- Enhances online privacy
- Protects sensitive information
- Prevents tracking and monitoring
- Increases browsing security
- Helps bypass geographical restrictions

Disadvantages of Anonymizers

- Can reduce internet speed
- Some services may store user data
- May be misused for cyber crimes
- Free anonymizers may not provide complete security

Prevention Against Misuse

- Monitor suspicious anonymous activities
- Use secure and trusted anonymizer services
- Follow cybersecurity laws and policies
- Maintain proper network monitoring systems

Phishing

PHISHING AND PHISHING ATTACKS

Don't Take the Bait. Stay Aware, Stay Secure.

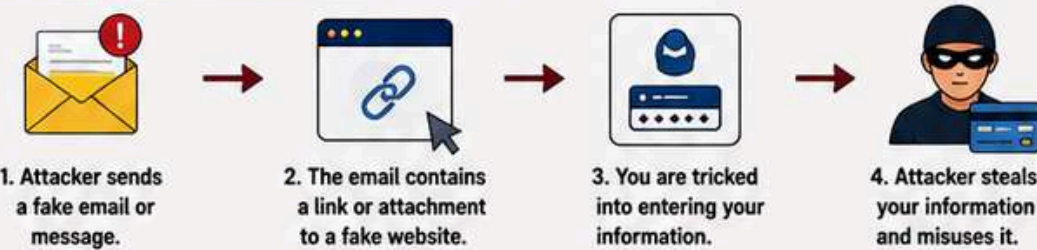
Phishing is a type of cyber attack where criminals pretend to be a trusted person or organization to trick you into revealing sensitive information such as usernames, passwords, credit card details, or OTPs. It is one of the most common and dangerous online threats.



WHAT IS PHISHING?

Phishing is a social engineering attack in which attackers send fake emails, messages, or create fake websites that look real, to steal your personal and financial information.

HOW PHISHING ATTACK WORKS



COMMON EXAMPLES

- Email Phishing**
Fake emails that look like they come from banks, companies, or trusted organizations.
- Spear Phishing**
Targeted attacks on specific individuals or organizations using personalized information.
- Smishing (SMS Phishing)**
Fake text messages that contain malicious links to steal your data.
- Vishing (Voice Phishing)**
Fraudulent phone calls that trick you into sharing sensitive information.
- Pharming**
Redirecting you to fake websites even if you type the correct URL.

SIGNS OF A PHISHING ATTACK

- Unexpected emails or messages
- Urgent or threatening language
- Requests for personal information
- Suspicious links or attachments
- Spelling and grammar mistakes
- Emails not addressed to you properly
- Sender's email address looks unusual
- Too good to be true offers or rewards



REAL-LIFE EXAMPLE

You receive an email that looks like it's from your bank. It says your account will be locked unless you verify your details. You click on the link, enter your username and password on the fake website, and later you find out your account has been hacked.

From: support@yourbank-secure.com
To: you@example.com
Subject: Urgent: Verify Your Account

Dear Customer,
We detected unusual activity in your account.
Please click the link below to verify your identity and avoid account suspension.

[Verify Now](#)

Thank you,
Your Bank Security Team



HOW TO PROTECT YOURSELF

- Don't click on suspicious links or attachments.
- Never share personal or financial information.
- Check the sender's email address carefully.
- Ensure the website uses HTTPS and is genuine.
- Use strong passwords and enable two-factor authentication.
- Keep your software and antivirus updated.
- Stay informed and aware about phishing tactics.

KEY TAKEAWAY

Phishing attacks rely on tricking you, not hacking your device. Always think before you click!
Be Alert. Be Safe. Be Secure.

WHAT TO DO IF YOU ARE A VICTIM

- Change your passwords immediately.
- Contact your bank or service provider.
- Report the incident to the authorities.
- Scan your device for malware.

REPORT PHISHING
Report suspicious emails to your email provider and report cyber crimes at www.cybercrime.gov.in

Password Cracking

Password cracking is the process of identifying or recovering passwords to gain unauthorized access to computer systems, networks, or online accounts. Cyber criminals use different techniques and software tools to guess or break passwords. Weak passwords are easier to crack and can lead to data theft, financial loss, and privacy breaches.

Common Methods of Password Cracking

- Brute force attack using multiple password combinations
- Dictionary attack using common words and passwords
- Phishing to trick users into revealing passwords
- Guessing passwords based on personal information
- Using malware or keyloggers to capture passwords

Prevention Measures

- Use strong and unique passwords

Password Cracking

- Enable two-factor authentication
- Change passwords regularly
- Avoid sharing passwords
- Use password managers

Keyloggers

A keylogger is a type of malicious software or hardware device that secretly records every keystroke typed on a keyboard. Cyber criminals use keyloggers to steal usernames, passwords, bank details, OTPs, and other confidential information.

- Keyloggers operate without the user's knowledge and send the collected information to attackers.

Types of Keyloggers

- Software keyloggers installed through malware
- Hardware keyloggers connected to computers physically

Effects of Keyloggers

- Theft of login credentials

Password Cracking

- Financial fraud
- Privacy violations
- Unauthorized account access

Prevention Measures

- Install antivirus and anti-malware software
- Avoid suspicious downloads and links
- Keep systems updated
- Use virtual keyboards for sensitive information
- Regularly scan devices for malware

Spyware

SPYWARE

The Silent Threat that Spies on You!

Spyware is a type of **malicious software** that secretly enters your device and collects your personal information without your knowledge. It monitors your activities and sends the data to cyber criminals.



WHAT DOES SPYWARE DO?

- Collects Personal Information**
Steals data such as usernames, passwords, emails, contacts, bank details, etc.
- Monitors User Activities**
Tracks browsing history, search queries, keystrokes, online activities and chat messages.
- Sends Information to Hackers**
The stolen data is sent to cyber criminals without your knowledge.
- Displays Unwanted Ads**
Shows pop-up ads and may change browser settings.
- Slows Down System**
Consumes system resources and reduces device performance.



EFFECTS OF SPYWARE

- Theft of personal and financial information
- Slow computer and poor performance
- Unauthorized access to accounts
- Privacy violations and identity theft
- Frequent pop-up ads and redirects

TYPES OF SPYWARE

- Adware**
Displays unwanted advertisements and tracks browsing activities.
- Keyloggers**
Records keystrokes to steal passwords and confidential information.
- Trojan Spyware**
Pretends to be useful software but secretly collects user information.
- Tracking Cookies**
Monitors browsing behavior and tracks online activities through cookies.

SIGNS OF INFECTION

- System becomes slow or freezes frequently
- Unknown programs appear automatically
- Excessive pop-up advertisements
- Browser homepage changes unexpectedly
- Increased internet usage

PREVENTION MEASURES

- Install antivirus and anti-spyware software.
- Avoid downloading files from untrusted websites.
- Do not click on suspicious links or attachments.
- Keep operating system and software updated.
- Use firewalls and secure browsers.
- Regularly scan devices for malware.

IMPORTANT!
Spyware can steal your identity, money and privacy. Always stay alert and follow safe computing practices.

SAFE COMPUTING TIPS

- Use strong passwords
- Enable two-factor authentication
- Do not share personal information online
- Backup important data regularly
- Be cautious while browsing internet



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